Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 1

Title: Retail Sales Analysis Using Power BI & MySQL

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Experiment No.: 1

Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

You are a Data Analyst in a retail company. The company stores its sales and customer-related data in flat files such as Excel/CSV and also maintains sales data in a MySQL database. Management wants a centralized dashboard to analyze retail performance. The objective is to connect these different data sources to Power BI Desktop, import the required data, explore and transform the data, and create interactive visualizations for analyzing sales by product category, store, customer, and time.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the retail sales dataset in CSV format.
- Obtained retail sales data from the MySQL database.
- Identified important fields such as Transaction ID, Date, Customer ID, Product Category, Quantity, Price per Unit, Total Amount, and Store.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported the CSV retail sales dataset using Get Data → Text/CSV.
- Created a retail_db database in MySQL.
- Created the sales table in MySQL.
- Inserted retail sales records into the MySQL table.
- Connected Power BI Desktop to the MySQL database and imported the sales table.

Step 3: Data Cleaning and Transformation

- Checked the imported data for errors and missing values.
- Verified the data types of columns such as Date, Quantity, Price per Unit, and Total Amount.
- Renamed fields where necessary.
- Ensured that numerical fields were correctly recognized for calculations.

Step 4: Power BI Interface Exploration

- Explored Report View for creating dashboards.
- Used the Data pane to select required fields.

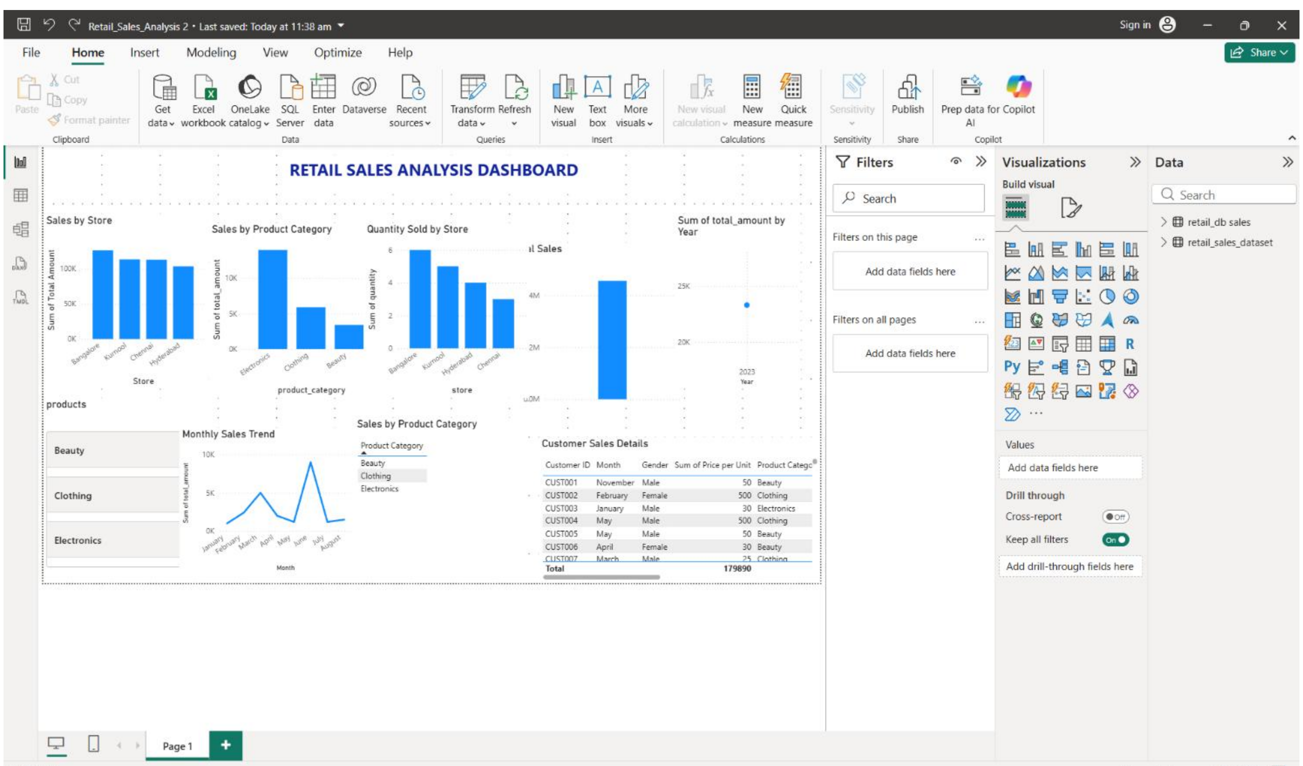
- Used the Visualizations pane to create and configure charts.
- Used filters to interact with the data.
- Worked with X-axis, Y-axis, legends, and visual formatting options.

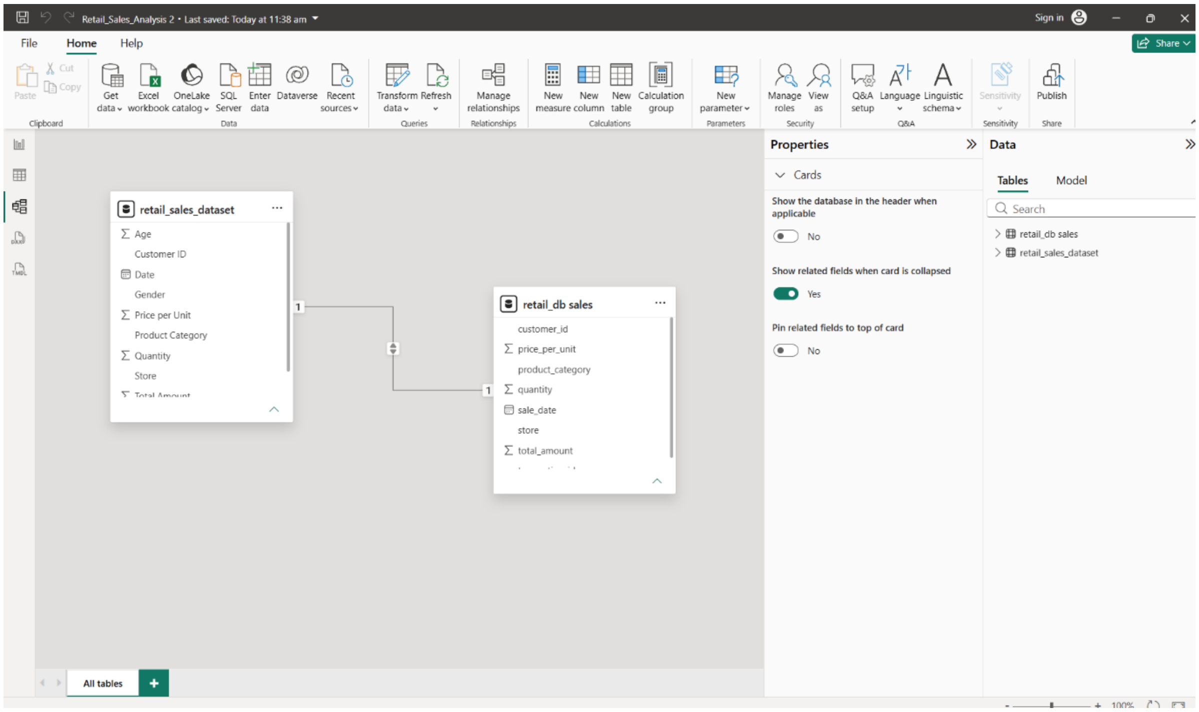
Step 5: Dashboard Development

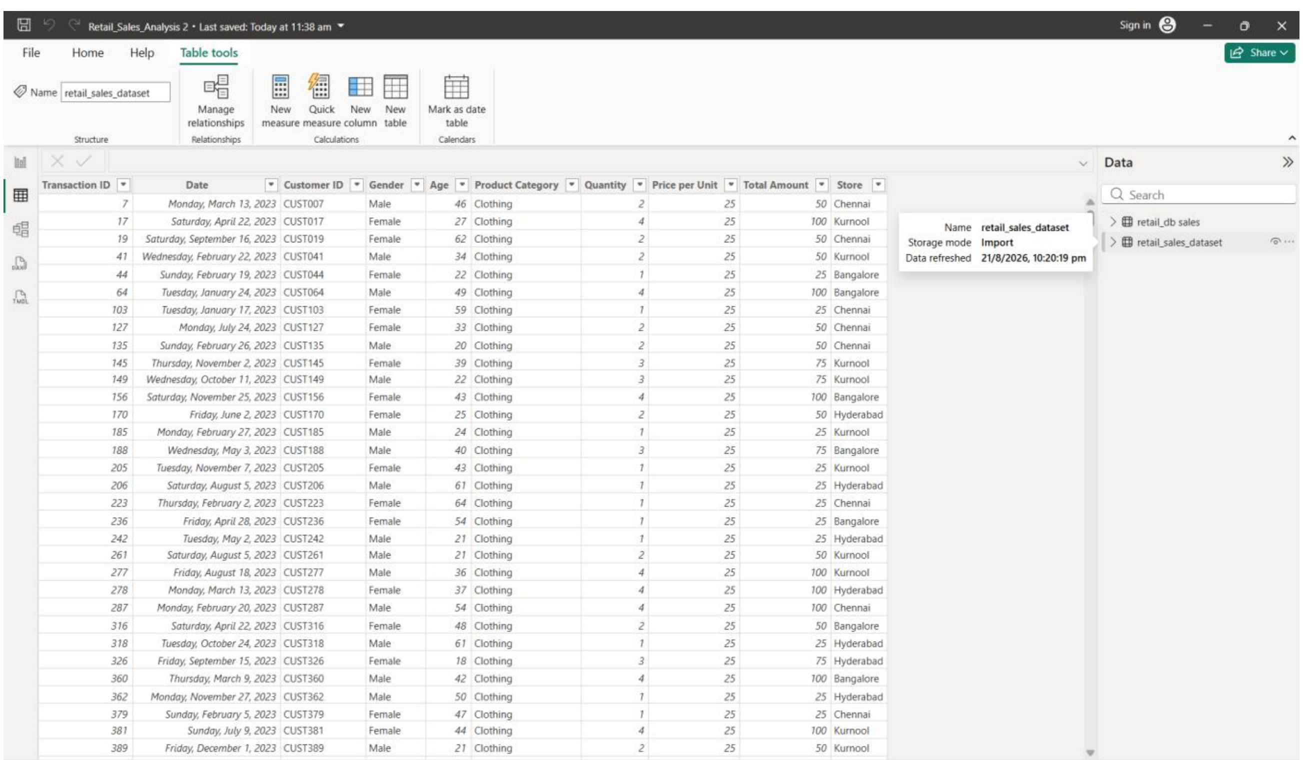
Created a Retail Sales Analysis Dashboard containing:

- Total Sales – KPI card showing overall sales.
- Sales by Store – Column chart comparing sales across stores.
- Sales by Product Category – Column chart comparing categories such as Beauty, Clothing, and Electronics.
- Sales Trend by Time – Time-based visualization for analyzing sales over time.
- Quantity Sold by Store – Chart showing the quantity of products sold by each store.
- Customer Sales Details – Table showing customer-related sales information.

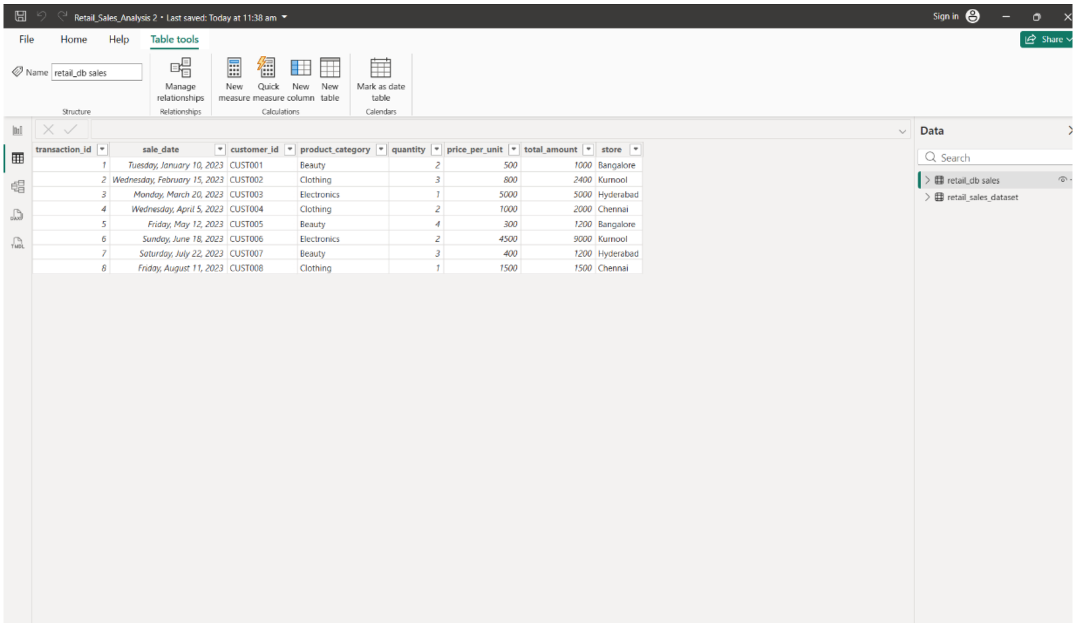
Output







Transaction ID	Date	Customer ID	Gender	Age	Product Category	Quantity	Price per Unit	Total Amount	Store
7	Monday, March 13, 2023	CUST007	Male	46	Clothing	2	25	50	Chennai
17	Saturday, April 22, 2023	CUST017	Female	27	Clothing	4	25	100	Kurnool
19	Saturday, September 16, 2023	CUST019	Female	62	Clothing	2	25	50	Chennai
41	Wednesday, February 22, 2023	CUST041	Male	34	Clothing	2	25	50	Kurnool
44	Sunday, February 19, 2023	CUST044	Female	22	Clothing	1	25	25	Bangalore
64	Tuesday, January 24, 2023	CUST064	Male	49	Clothing	4	25	100	Bangalore
103	Tuesday, January 17, 2023	CUST103	Female	59	Clothing	1	25	25	Chennai
127	Monday, July 24, 2023	CUST127	Female	33	Clothing	2	25	50	Chennai
135	Sunday, February 26, 2023	CUST135	Male	20	Clothing	2	25	50	Chennai
145	Thursday, November 2, 2023	CUST145	Female	39	Clothing	3	25	75	Kurnool
149	Wednesday, October 11, 2023	CUST149	Male	22	Clothing	3	25	75	Kurnool
156	Saturday, November 25, 2023	CUST156	Female	43	Clothing	4	25	100	Bangalore
170	Friday, June 2, 2023	CUST170	Female	25	Clothing	2	25	50	Hyderabad
185	Monday, February 27, 2023	CUST185	Male	24	Clothing	1	25	25	Kurnool
188	Wednesday, May 3, 2023	CUST188	Male	40	Clothing	3	25	75	Bangalore
205	Tuesday, November 7, 2023	CUST205	Female	43	Clothing	1	25	25	Kurnool
206	Saturday, August 5, 2023	CUST206	Male	61	Clothing	1	25	25	Hyderabad
223	Thursday, February 2, 2023	CUST223	Female	64	Clothing	1	25	25	Chennai
236	Friday, April 28, 2023	CUST236	Female	54	Clothing	1	25	25	Bangalore
242	Tuesday, May 2, 2023	CUST242	Male	21	Clothing	1	25	25	Hyderabad
261	Saturday, August 5, 2023	CUST261	Male	21	Clothing	2	25	50	Kurnool
277	Friday, August 18, 2023	CUST277	Male	36	Clothing	4	25	100	Kurnool
278	Monday, March 13, 2023	CUST278	Female	37	Clothing	4	25	100	Hyderabad
287	Monday, February 20, 2023	CUST287	Male	54	Clothing	4	25	100	Chennai
316	Saturday, April 22, 2023	CUST316	Female	48	Clothing	2	25	50	Bangalore
318	Tuesday, October 24, 2023	CUST318	Male	61	Clothing	1	25	25	Hyderabad
326	Friday, September 15, 2023	CUST326	Female	18	Clothing	3	25	75	Hyderabad
360	Thursday, March 9, 2023	CUST360	Male	42	Clothing	4	25	100	Bangalore
362	Monday, November 27, 2023	CUST362	Male	50	Clothing	1	25	25	Hyderabad
379	Sunday, February 5, 2023	CUST379	Female	47	Clothing	1	25	25	Chennai
381	Sunday, July 9, 2023	CUST381	Female	44	Clothing	4	25	100	Kurnool
389	Friday, December 1, 2023	CUST389	Male	21	Clothing	2	25	50	Kurnool



transaction_id	sale_date	customer_id	product_category	quantity	price_per_unit	total_amount	store
1	Tuesday, January 10, 2023	CUST001	Beauty	2	500	1000	Bangalore
2	Wednesday, February 15, 2023	CUST002	Clothing	3	800	2400	Kurnool
3	Monday, March 20, 2023	CUST003	Electronics	1	5000	5000	Hyderabad
4	Wednesday, April 5, 2023	CUST004	Clothing	2	1000	2000	Chennai
5	Friday, May 12, 2023	CUST005	Beauty	4	300	1200	Bangalore
6	Sunday, June 18, 2023	CUST006	Electronics	2	4500	9000	Kurnool
7	Saturday, July 22, 2023	CUST007	Beauty	3	400	1200	Hyderabad
8	Friday, August 11, 2023	CUST008	Clothing	1	1500	1500	Chennai

Explanation of Data in the Dashboard

The dashboard represents retail sales data and provides an overview of sales performance based on stores, product categories, quantities sold, customers, and months. The data is connected to Power BI and presented using different charts and tables for easy analysis.

- Total Sales: This shows the overall sales amount generated from all transactions in the dataset. It provides a quick view of the business's total revenue.
- Sales by Store: This chart compares sales across different stores such as Bangalore, Kurnool, Chennai, and Hyderabad. It helps identify which store contributes the most to total sales.
- Sales by Product Category: This visual compares sales across Electronics, Clothing, and Beauty. From the dashboard, Electronics has the highest sales among the categories.
- Sum of Total Amount by Year: This visual presents the total sales amount for the available year, helping understand the yearly sales performance.
- Quantity Sold by Store: This chart shows the total quantity of products sold by each store. Bangalore has the highest quantity sold, followed by Kurnool, Hyderabad, and Chennai.
- Customer Sales Details: This table provides detailed transaction-level information such as Customer ID, Month, Gender, Price per Unit, and Product Category. It allows individual customer transactions to be examined.
- Monthly Sales Trend: The line chart shows how sales change from month to month. It helps identify periods of higher and lower sales and understand the overall sales pattern during the year.
- Product Category Filter: The dashboard includes a filter for Beauty, Clothing, and Electronics, allowing users to select a specific category and analyze its sales information.

The final Power BI dashboard provides:

- Total Sales

Sales by Store

- Sales by Product Category
- Sales Trend by Year
- Quantity Sold by Store
- Customer Sales Details
- Sales Trend by Month

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Conclusion

The retail sales data was successfully connected to Power BI Desktop from both CSV and MySQL data sources. The data was imported, explored, and analyzed using Power BI visualizations. A dashboard was created to analyze total sales, sales by store, product category, customer, and time. The final dashboard provides a clear and interactive view of sales performance and helps in understanding important retail business trends.

Learning Outcomes

Through this project, I learned:

- Learned how to connect Power BI Desktop to CSV/Excel and MySQL data sources.
- Learned how to import and organize data in Power BI.
- Understood the basic Power BI interface, including Report View, Data View, Data pane, and Visualizations pane.
- Learned how to create and format charts, tables, and KPI cards.
- Learned how to analyze sales by product category, store, customer, and time.
- Gained practical knowledge of data filtering and visualization.
- Learned how to create an interactive retail sales dashboard for business analysis.
- Developed basic skills in presenting data-driven insights using Power BI.